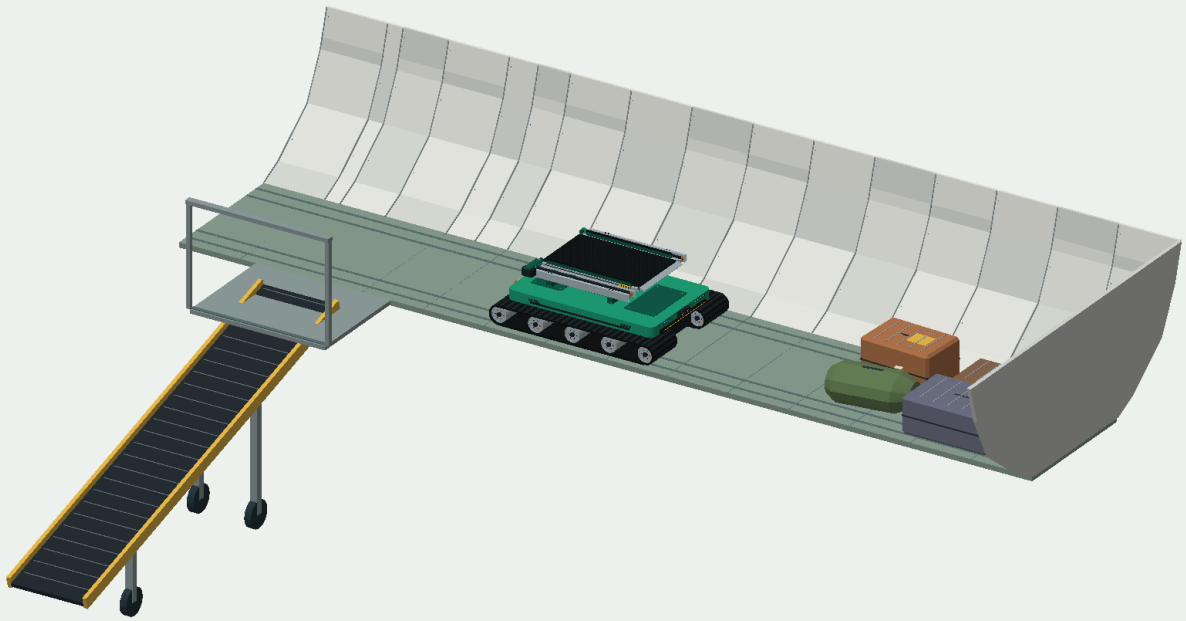
BELLYBOT
Aircraft environment & fit review

VERSION 1.0 / 10 OCTOBER 2026 / INTEGRATION REVIEW

BELLYBOT / AIRCRAFT ENVIRONMENT

7.70 m forward-hold fixture / 1.12 m ceiling / side door and apron belt loader



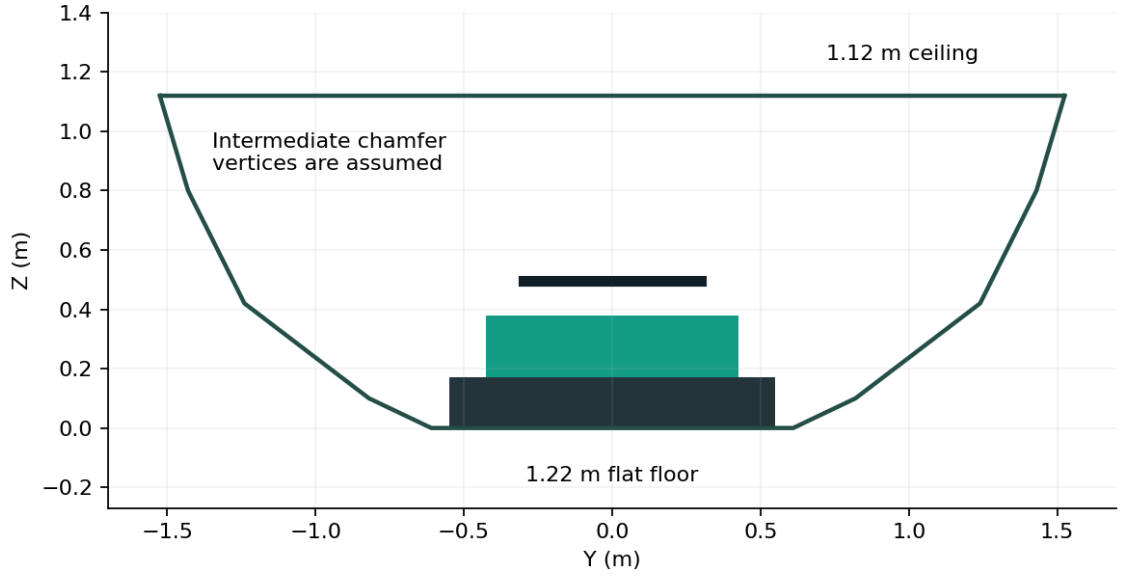
Actual delivered meshes | Reference-based test geometry; intermediate dimensions are assumptions

A confined baggage-handling test environment built from original meshes, the supplied reference photos and a published narrow-body dimensional baseline. The robot shown is the existing BellyBot v3 geometry. Roof and near-side wall are removed in this illustration; simulation collision boundaries remain present.

Deliverable	Purpose
5 worlds / 10 reusable models	Manual-reference scene, enclosed robot tests, contact belt loader, varied free bags and an aft-taper stress case.
SDF / URDF / meshes / STEP / source	Gazebo integration, static RViz inspection, CAD layout and reproducible geometry.
Team details	Team, lead, members and institute: complete in the final hackathon submission.

Status: offline checks completed; Gazebo and RViz were not available to run. Door-entry routing, the corner conveyor handoff and autonomous loading remain integration work. This document supplements the robot report.

01 / Reference basis and geometric envelope



Dimension	Baseline / provenance
Forward test hold	7.70 m length; 1.22 m flat floor; 3.05 m upper width; 1.12 m height. Published Boeing 737 example [2].
Side door	1.22 m width along the hold and 0.89 m clear height [2]. Door station, local tunnel and 8 mm sill are assumed.
Interior detailing	Faceted liners, rails, lights and tan restraint material reflect the supplied images. Dimensions of these details are assumed.
Aft stress fixture	10.9 m length; ceiling tapers from 1.19 to 0.59 m and upper width from 2.92 to 2.08 m. Endpoints are sourced; linear interpolation is assumed.

The photos do not identify an aircraft variant or provide a calibrated survey. This fixture is a 737-inspired baseline, not an as-built reconstruction. All collision and visual dimensions can be changed in **config/environment.json**. The origin is the near bulkhead on the cargo floor; +X points along the hold, +Y away from the door and +Z upward.

The video link resolves to Tim Falagan's *Ramp Agent.. “A” Team!*, 139 seconds. Playback and frame retrieval were unavailable. No video timestamps, measured handling rates or motion observations are invented; the manual scene uses the three supplied stills.

02 / Manual handling and the robotic task

INTERIOR / LOW-CEILING MANUAL WORKSPACE



The supplied images show two workers kneeling or leaning forward beneath a low ceiling, with a suitcase handled near the side opening. The scene recreates that restricted workspace using fixed human proxies. It is a posture illustration and occupancy obstacle scene, not an ergonomic risk score or a human dynamics simulation.

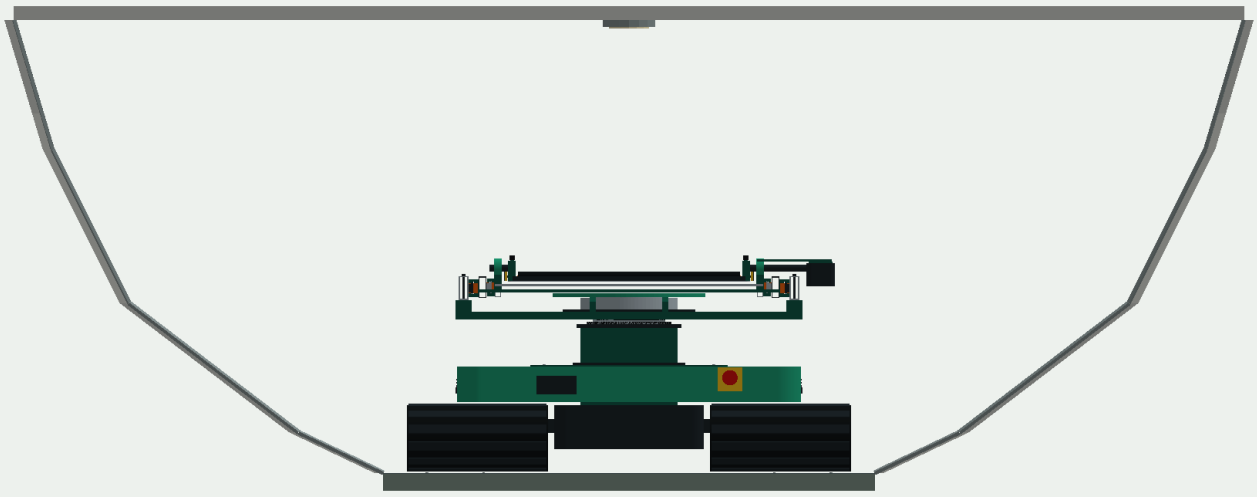
Task stage	What the integration should evaluate
Receive at door	Keep a supported bag path over the sill; detect jammed or badly oriented luggage.
Transfer and orient	Replace the implied manual redirection with an engineered 90-degree, height-matched transfer into the longitudinal infeed.
Place / retrieve	Use the robot tray within a checked free-space envelope; confirm stable support before release. Keep operators outside the active zone.

The environment supplies free rigid bags and an external contact-driven conveyor. Soft-bag deformation, straps, real garment contact and injury outcomes are not modeled. A scene containing people does not automatically stop the robot: detection and common stop supervision must be implemented.

03 / BellyBot fit: what works and what conflicts

CROSS SECTION / STOWED BELLYBOT

1.22 m flat floor / 1.10 m crawler width / no full chassis pivot assumed

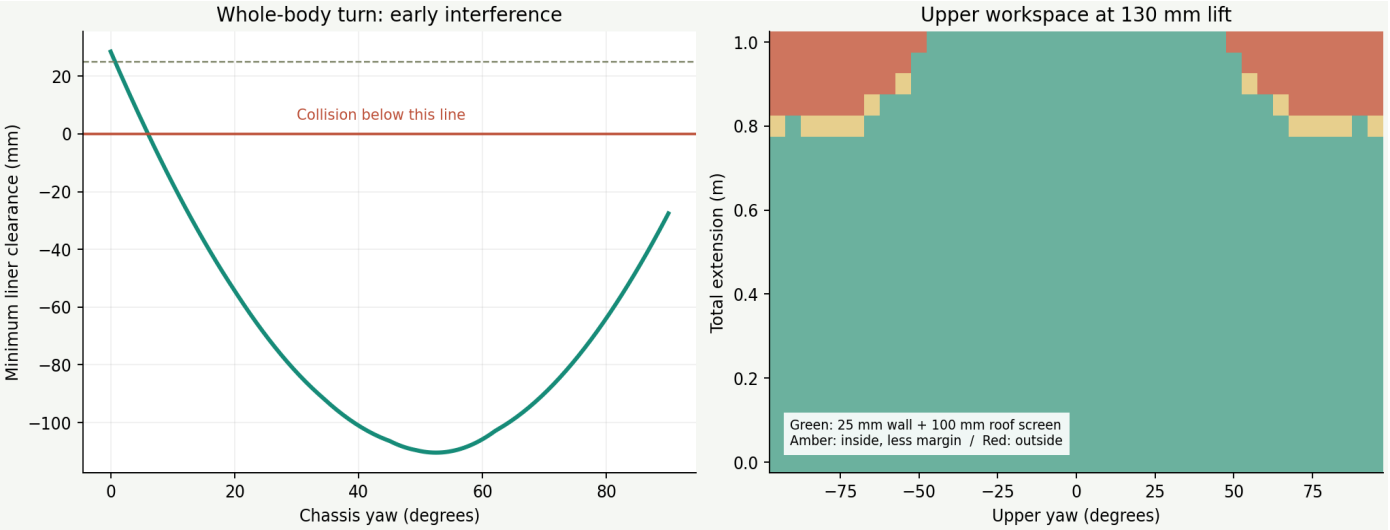


Actual delivered meshes | Reference-based test geometry; intermediate dimensions are assumptions

Check	Result in the forward fixture
Stowed crawler	1.10 m wide on a 1.22 m floor. Nominal nearest sloped-liner clearance: 28.5 mm at the 3 mm raised spawn.
Whole-base pivot	First sampled liner intersection at 6.5 degrees; 25 mm margin fails at 1 degree. Do not enable chassis pivoting in the work zone.
Maximum lift	600 mm lift reaches 1.138 m including spawn offset: 18 mm beyond the ceiling.
Useful loaded operation	A 380 mm-high bag at 130 mm lift and a 280 mm-high bag at 230 mm lift retain about 110 mm roof clearance.

The upper yaw joint is distinct from chassis yaw. The checked 65-degree upper-yaw placement and 95-degree / 600 mm extension pose fit the assumed enclosure. A centered robot is already close to the 25 mm wall-margin screen; geometric uncertainty and lateral tracking error matter.

04 / Operating limits and required design changes



The upper-workspace grid uses a fixed 130 mm lift and a conservative 780 × 500 × 320 mm bag envelope. Green indicates the selected 25 mm wall and 100 mm roof margins. It does not account for an evolving baggage pile, other mechanisms or all continuous trajectories. Use it as evidence for a pose-dependent guard, not a universal joint-limit table.

Priority	Recommended action
1. Adapt the controller	Set the forward local roof to 1.12 m. Cap total lift from measured bag height; use 130 mm for an unknown bag up to 380 mm. Match the infeed height. Keep base travel longitudinal.
2. Resolve entry and handoff	Prove a full door/sill entry path or design modular installation. The sampled door pivot fails in 36 of 46 poses. Engineer the side-to-longitudinal transfer; a gap between conveyors is not a loading mechanism.
3. Preserve contact and stability	Do not shrink the base without recomputing stability and contact pressure. Prior v3 peak screening is 594.31 kg/m² against a 600 limit, with unvalidated mass/contact assumptions.
4. Reduce avoidable cycle time	Batch by layer height and orientation; avoid a full chassis turn or floor-pitch cycle per bag. Measure completed placements, recovery and interruptions before claiming 12 bags/min.

The aft stress test also rejects loaded stowed travel in its lower far region. A lower receiving tray, restricted work zone or revised transfer method would be needed there. The environment does not imply a load-capacity or hardware safety rating.

05 / Integration, evidence and hackathon readiness

Evidence	Status
XML and resource checks	15 SDF files; 632 local references resolved; original meshes are self-contained.
Mesh / inertial checks	600 closed collision meshes; positive physical inertia tensors for dynamic models; cutaway keeps the full collision set.
Robot enclosure sampling	1066 sampled poses; known failures are reported, not suppressed. No continuous global path proof.
CAD	Forward fixture STEP reimported as 172 valid solids; millimetres. SDF, URDF and mesh coordinates are metres.
Runtime / performance	Gazebo, RViz, bag contact, sustained throughput and structural floor loading: not run / not demonstrated.

Handoff: read **docs/AGENT_HANDOFF.md**; launch **robot_clearance**; compose the existing v3 SDF into a new world; inspect and settle; enforce the proposed limits; verify individual joints and one free-bag transfer; then run the logged loading task. The compose tool preserves the existing robot package and assumes pre-positioning inside the hold.

Score correctly: use simulation time and unique bag IDs. A placement is accepted only after stable support and tray clearance. The scorer excludes duplicate placements and dropped cases. Supply summed normal force and actual non-overlapping patch area for local floor loading. Neither a bag spawn interval nor whole-floor average demonstrates the hackathon target.

Submission mapping: the supplied guideline requires STEP, a maximum two-minute video and a 5–8 page concept report with team details. This ZIP supplies an environment CAD layout and this six-page review. Update the existing robot animation/report after integration and use the prescribed team filenames. No final submission or compliance claim has been made.

Sources

[1] Supplied Hackathon Guidelines, Oct 2, and three user photographs (included in reference/).
[2] Boeing, 737NG passenger overview, PDF p.17:
boeing.com/content/dam/boeing/v2/company/startupboeing/historical/737NG_passenger.pdf.
[3] User video: youtube.com/watch?v=6gra6c2N8n4 (metadata only).
[4] Gazebo Harmonic ROS installation and Sim 8 TrackController API. Full links, assumptions and limitations: docs/REFERENCES.md.